SINGAPORE UNIVERSITY OF TECHNOLOGY AND DESIGN

Singapore University of Technology and Design

50.043 Database Systems

Group 2

SimpleDB - Lab4

Authors	***Student ID***
Jone Chong Jin	1006338
Sarang Nambiar	1006181
Loh Jianyang John	1006360

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Implementation Details:

1. Explain any design decisions you made.

<u>BufferPool.java (lab 2 implementation revisit)</u>

A small but important adjustment was made in `BufferPool.insertTuple()`, `BufferPool.flushPage()` and `BufferPool.deleteTuple()`:
Instead of assuming all tables use `HeapFile`, we modified the method to operate on the generic `DbFile` interface.

We needed to make the above changes in order to support insertion and deletion from both `HeapFile` and `BTreeFile` objects introduced in Lab 4, and to avoid `ClassCastException` errors during B+ tree deletion and locking tests.

Exercise 1, 2, 3 and 4:

The implementations for Exercises 1, 2, 3 and 4 follow the standard procedures described in the lab instructions. No additional design decisions beyond the provided specifications were made.

2. Explain the non-trivial part of your code.

Exercise 1:

In the `findLeafPage` function, we iterate through all the keys of the current internal node, comparing if they are greater than or equal than our provided key. Only when none of the keys are greater than or equal than our provided key do we recursively go down the right side of the tree. This logic aligns with the algorithm as it prevents us from going down the right side of the tree prematurely.

Exercise 2:

The `splitLeaf` and `splitInternalPage` functions both compare the key to be inserted with the key of the middle entry in the leaf or internal page. This comparison checks whether the middle key is greater than or equal to the key being inserted. This logic aligns with the search algorithm, which always explores the left page first when a key is lesser than or equal to the middle key.

3. **Discuss and justify any changes you made to the API.**
   No changes were made to the original API for the implementation of this task.

4. **Describe any missing or incomplete elements of your code.**
   None.